

Referee audit of “ $\zeta(5)$ is irrational” (A. Fauzan, 17 September 2026)

Claude Opus 5 (1M context), Anthropic
an AI system, run as an agentic workflow in Claude Code
prepared for Dan Romik

22 September 2026

1 Executive summary

What the paper claims

The preprint — Aabir Fauzan, *$\zeta(5)$ is irrational*, 17 September 2026, Zenodo, doi:10.5281/zenodo.22826418; this audit is of version v1, doi:10.5281/zenodo.22826419, and all page and equation references are to that version (32 pp., single author) — proves two things. **Theorem 1.1:** $\zeta(5)$ is irrational. **Corollary 1.2:** $|\zeta(5) - a/b| > b^{-260}$ for all integers a and all $b \geq b_0$, with b_0 unspecified. Corollary 1.2 is not needed for Theorem 1.1.

How the proof is organised

Fix $K = 40n$, $N = 3n$, $h = 37n$, $\alpha = 3/40$, $\lambda = 37/40$, and a cutoff $M \geq 40$ with $K \geq 200M^2$. Equations (2.2)–(2.3) define a $\mathbb{Q}$ -linear functional μ_X on rational functions with simple poles among the $-j^2$, with values in $\mathbb{Q}[X]$ and affine dependence on X . Proposition 2.2 shows that at $X = \zeta(5)$ the functional is integration against the positive weight $w(y) = \frac{(2\pi)^4}{12} y^5 \sum_{\ell \geq 1} \ell^4 e^{-2\pi \ell y}$, so the $h \times h$ Hankel matrix $G_K(X) = [\mu_X(D_N^6 t^{i+j}/D_K)]$ is at $X = \zeta(5)$ a positive moment matrix; (2.9) computes the leading coefficient of $\Delta_K = \det G_K$ by a Vandermonde identity and shows $\deg \Delta_K = h$ exactly. Sections 3–5 clear denominators: a p -adic extension of μ_X to a Tate algebra (Lemma 3.1), a distribution relation (Lemma 3.2) and a crude small-prime bound (Lemma 3.3) yield local determinant valuations γ_p^{in} (Proposition 4.1, for $K/M < p \leq K/3$) and γ_p^{out} (Proposition 4.3, for $p > K/3$); (5.1)–(5.3) assemble these into an explicit positive rational $m_{K,M}$ with $Q_{K,M} = m_{K,M} S_K \Delta_K \in \mathbb{Z}[X]$ (Proposition 5.1), and the prime number theorem converts the local bounds into $\limsup K^{-2} \log m_{K,M} \leq A_M$ (Proposition 5.2). Section 6 bounds the real side: Andréief’s identity (6.10) writes $\Delta_K(\zeta(5))$ as an h -fold integral with a squared Vandermonde, which after the scaling $y_i = K \sqrt{t_i}$ is a discrete logarithmic energy in the external field V of (6.1); comparison with an explicit sum of sixteen arcsine measures (Lemma 6.1) together with negativity of zero-mass logarithmic energy (Lemma 6.2) gives $\log F_K(\zeta(5)) \leq \overline{U} K^2 + 24K \log K + 200K$ (Proposition 6.3). Section 7 adds the two constants: $A_{200} + \overline{U} < -139/8000 < 0$, which is (2.7); then $b^{37n} Q_n(a/b)$ is a positive integer tending to zero, and Theorem 1.1 follows.

The verdict

Verdict. We found no error. Nine independent audit tracks, each followed by a separate adversarial verification pass, checked every numbered statement of the paper that reduces to a computation — most of them in exact rational or p -adic arithmetic, several at 25–6000 significant digits. Not one of the paper’s statements came out false. The claimed identities that can be checked exactly are exact; the claimed inequalities that can be checked hold, several of them with equality; and the two decisive constants reproduce digit for digit from the paper’s own data.

The paper as written is not a complete proof. It has one gap: on p. 10 the inequality $b_a \leq 6\alpha x + 3$ is asserted without proof and attributed to (4.4), which is not its source. The gap is load-bearing — with the naive bound $6\alpha x + 6$ the dimensions L_a go negative at $x = 3$ and Proposition 4.1 collapses — but it is small: we supply the missing two lines in Finding 1, and with them the argument goes through. By the ordinary standard, a proof with an unproved load-bearing step is not correct, so the paper needs that correction before it can be called one. It is a repairable defect, not a false statement and not an unproved assumption carrying the theorem.

With that gap filled, we found no further error. We read every proof in the paper and reconstructed the ones that carry weight; we found each of them complete. We could not, however, *certify* §4.1: two careful readings of a dense p -adic valuation argument are evidence, not refereeing, and unlike every other estimate in the paper its conclusion cannot be cross-checked against a computed determinant, because its own hypothesis (4.1) forces a matrix of size ≥ 296000 . That is a limitation of our cross-checking, not a defect in the paper’s argument, and we say so plainly here because an earlier draft of this report phrased it in a way that suggested otherwise.

So the verdict is: *one repairable gap, which we have repaired; no other error found; and one section whose proof we have read and believe but cannot certify.* What this is not is a certification. “No error found” is not “no error”, and the distinction matters most where we could not compute — if §4 over-claims γ_p^{in} , the *proof* fails while the *theorem* may well survive, for the reason given in Section 5.

Confidence, and what would change it

We separate three questions, because our confidence in them differs.

1. *Do Sections 1–3, 5–7 and Appendices A–C contain an error?* **Very high confidence that they do not** (≈ 0.95). Everything in this range was verified independently, most of it twice by different methods; Appendix A’s certification of Lemma 6.1, which the audit initially recorded as uncheckable, was in the end fully reproduced (Section 3.7), and Appendix B reproduces exactly as rationals. The figure is set by the weakest component rather than the strongest: the verifying pass on §5 and Appendix B put itself at 0.97 (conditional on (3.12), (4.8) and (4.14)), the pass on §6 and Appendix A at 0.95, and a conjunction cannot beat its weakest term. What is left in this range is reading, not computing — the proofs of Lemmas 3.1–3.3, 4.2 and 6.2 and the partial summation in Proposition 5.2 were audited line by line and machine-tested only through their consequences.
2. *Is Section 4 correct?* **High confidence** (≈ 0.9). The outer range (Proposition 4.3) is verified against exact determinants at 37 primes with *equality at 27 of them*, which is very

strong evidence for a formula of this complexity. The inner range (Proposition 4.1) is proved in the paper (pp. 10–11), and we reconstructed that proof line by line, twice and independently, and found it complete once the missing inequality of Finding 1 is inserted. What we could not do is check its conclusion against a computed determinant, since (4.1) forces $p > 8000$ and a matrix of size ≥ 296000 ; instead its mechanism was tested at the smallest nondegenerate instance ($K = 80$, $p = 23$), where it reproduces the true valuation $v_{23}^G(\Delta_{80}) = -188$ *exactly*, and where switching off the paper’s side conditions makes the bound false. All side conditions were swept over tens of thousands of primes at the real parameters with no failure.

3. *Is the theorem true?* **High confidence** (≈ 0.95), and this is worth separating from (2), because the evidence is of a different kind. By the paper’s own remark C.3, irrationality follows from $\limsup K^{-2} \log P_K(\zeta(5)) < 0$ for the *primitive* polynomials $P_K = \Delta_K / \text{cont}(\Delta_K)$, with no reference to $m_{K,M}$ at all. Measured directly from exact determinants, $K^{-2} \log P_K(\zeta(5)) = -0.1657, -0.1302, -0.1253$ at $K = 40, 80, 120$, against a requirement of < -0.0174 : the construction has roughly seven times the room it needs. An error in Sections 3–5 would therefore damage the *proof* without necessarily touching the *theorem*.

Overall we would put the probability that the paper is a correct proof of Theorem 1.1 at roughly 0.85, and we would not be surprised to be told that a referee with more time found a repairable gap in Section 4. We would be surprised to be told the paper is wrong.

What would change the verdict. (i) A demonstration that $v_p^G(\Delta_K) < \gamma_p^{\text{in}}$ for some admissible inner prime — this is the one inequality nobody has tested at the paper’s parameters, and it would be fatal. (ii) An error in Lemma 3.1 or 3.2 of one power of p , which would move γ_p^{in} by about h and change A_M . (iii) Any correction to A_{200} or $\bar{U}$ exceeding 1.74×10^{-2} (1.29% of A_{200}), which is what it takes to reach Theorem 1.1. Three smaller thresholds sit below it and touch only the printed constants: 5.07×10^{-6} breaks (2.8) and with it Corollary 1.2, 3.28×10^{-5} forces the printed exponent 139/5 in (2.7) to be reprinted, and neither disturbs Theorem 1.1. Nothing we found moves either constant at all.

On the prior. The problem is old and famous and the author is unknown to us; the paper carries a disclosure that a generative AI tool was used for “consistency checks of the arguments, calculations, and references”. None of this is an argument. We record that the paper passes a test that no wrong paper in this circle of ideas has passed in our experience: its machinery reproduces $\zeta(3)$ (with a weak exponent ≈ 14.5) and *fails* for $\zeta(7)$, $\zeta(9)$, $\zeta(11)$, with balances $+0.93, +1.97, +3.04$ against -0.049 for $\zeta(5)$. It does not prove too much.

2 Summary table of the numbered statements

Verdicts: correct = checked and correct; imprecise = correct but imprecise, terse or not reproducible as written; **gap** = a gap in the written argument; **false** = false; not checked = could not check. “Exact” means exact rational or p -adic arithmetic.

Statement	Verdict	Comment
Section 1		
Theorem 1.1	correct	Deduction from Theorem 2.1 is airtight: quadratic decay beats linear height growth, so no height estimate is needed. Its truth is that of Theorem 2.1.
Corollary 1.2	imprecise	Argument valid, constants exact. But $b_0 > \exp(1.07 \times 10^{11})$: the <i>threshold</i> is not merely unspecified but ineffective, so the exponent 260 is not usable for any b one could write down. The paper says only “not specified”.
$0 < \zeta(5) - p/q < q^{-0.86}$	correct	Brown–Zudilin citation verified against arXiv:2210.03391 (lines 86, 180).
Section 2		
(2.1) parameters	correct	$1 + 2\alpha = 23/20$, $h = K - N = 37n$, $\lambda = 37/40$, $\alpha = 3/40$ all consistent.
(2.2) $\mu_X(t^e)$	correct	Symbolically identical to $(2e+5)! \zeta(2e+2)/(12(2\pi)^{2e+2})$ for $e = 0..11$; matches $\int y^{2e} w$ to 24–40 digits.
(2.3) $\mu_X(1/(t+j^2))$	correct	Equals $a^4 \zeta(5, a) - \frac{1}{2a} - \frac{1}{4}$ at $a = j$; both correction terms and both signs verified, symbolically and to ~ 24 digits, $j = 1..6$.
§2.1 extension of μ_X	imprecise	No circularity: $\{t^e\} \cup \{1/(t+j^2)\}$ is a basis, so (2.2)–(2.3) <i>define</i> rather than extend. Exclusion of $t = 0$ is used but never stated.
(2.4)–(2.6)	correct	G_K genuinely Hankel; $S_K > 0$; $m_{K,M} > 0$ by construction. S_K is independently forced by (3.11).
Theorem 2.1	not checked	Degree and positivity verified. Integrality is Proposition 5.1; decay is (7.2).
(2.7), (2.8)	correct	Follow from (7.2). Both constants are saturated: see Section 4.
(2.9) degree	correct	Exact identity, sign included, at $(N, K) = (1, 3), (2, 5), (3, 7), (1, 5), (3, 9)$ and at the real $(3, 40)$, where it matches a 1594-digit integer. Hence $\deg Q_{K,M} = h$ exactly.
Proposition 2.2	correct	Every step checked: $w = y^5 f^{(4)}/12$, $w(0^+) = 1/\pi$ to 20 digits; the quartic derivative identity symbolically; all four boundary terms $O(y)$; Hermite (DLMF 25.11.29) to ≥ 24 digits at five values of a . Only omission: the $\sum / \int$ interchange (Tonelli, one line).
(2.10)	correct	Positive definiteness of $G_K(\zeta(5))$ confirmed at $K = 40, 80, 120, 160, 200, 240$; at $K = 40$, all 37 LDL pivots positive, certified five ways including interval arithmetic.
Section 3		
§3.1 defs, κ_d , $\ \tau\ $	correct	$\ \tau\ \leq p$ is the correct power and is sharp (attained at z^{4p-1}). $B_1 = -1/2$ is stated (p. 3).
(3.1) pullback	correct	Exact on monomials ($e = 0..10$) and on poles ($j = 1..12$); reproduces (2.2) and (2.3).
(3.2), (3.3)	correct	From $B_n(1) = (-1)^n B_n$ and $B_n(x+1) - B_n(x) = nx^{n-1}$. The hypothesis “ g regular at zero” is necessary and consistent.

continued on the next page

Statement	Verdict	Comment
Lemma (3.4)	3.1, correct	Valid; tested exactly at $p = 7, 11, 13$. $d(r_\nu) < p$ is necessary ($r = p$ gives $v_p = -5$); $\deg U_0 \leq p + 1$ is sufficient but not tight (true threshold $4p - 2$); $r_\nu - r_\eta \in \mathbb{Z}_p^\times$ is redundant given $d(r_\nu) < p$ (now proved, not just searched).
(3.5)	correct	Exact; converges in the Tate algebra iff $v_p(s) < 0$.
far/near consistency (p. 7)	correct	True, and the paper's one-sentence justification (uniqueness of the B_T decomposition) is the right one. Verified on 18 random instances.
Lemma (3.6), (3.7)	3.2, correct	p^{-4} is right (p^{-1} Raabe, p^{-3} three derivatives); $Y = p^5 X + C_p$ is exactly what makes the $1/x$ case return $-X$. $C_p \in p^5 \mathbb{Z}_p$ for every $p \geq 7$ (a theorem, not a table); " $p \geq 7$ " is genuinely needed.
Lemma (3.8)–(3.10)	3.3, correct	Every estimate checks; (3.8) and (3.9) are sharp (min slack 0). (3.10) never violated in $180 + 216$ exact (K, d, p) cases on two different grids. Residue sign left as " $\pm$ "; it is $(-1)^{K-r}$.
(3.11)	correct	<i>The key "no lost power of p" test.</i> An exact identity: $\det[(K!)^2 \mu_X(f_i f_j / D_K)] = S_K \Delta_K$, with S_K precisely (2.5). Verified exactly at five (K, N, h) and then proved.
(3.12)	correct	Follows from (3.10) plus the ultrametric determinant bound; holds at every prime $< 5K + 200$ at $K = 40, 80, 120$ with large slack.
Section 4		
(4.1)	correct	Implies $p > 200M \geq 8000$ and $p^2 > 5K$; the latter <i>is</i> used, on p. 14 in §5.1.
(4.2), (4.3)	correct	Exact p -power extractions, not bounds. The " -4 " and " $+1$ " silently absorb the p^{-4} prefactor of (3.7).
(4.4), ordering rule	correct	$L_0 + \sum_a L_a = h$ is an identity. The rule "extras to the largest ℓ_K " is not cosmetic: it excludes the one bad case of the weight comparison.
$b_a \leq 6\alpha x + 3$ (p. 10)	gap	True and sharp, but asserted without proof and mis-attributed to (4.4). Needs a case split on $v_N = N \bmod p$. With the naive $6\alpha x + 6$ the chain gives $T - b_a > -3/2$ at $x = 3$ and $L_a \geq 0$ fails. See Finding 1.
(4.5) basis	correct	$\mathbb{Z}_p$ -unimodular by CRT; verified numerically ($v_p(\det U) = 0$).
(4.6)–(4.8)	correct	Re-derived; $\gamma_p^{\text{in}} \in \mathbb{Z}$; equals the true valuation exactly in the one computable nondegenerate instance.
Proposition 4.1	not checked	Proof reconstructed line by line and found complete. All three side conditions are displayed on pp. 10–11 (not hidden). <i>Not testable at its own parameters</i> ($h \geq 296000$); the $m_N \geq 1$ sub-range has no determinant test.
(4.9)	correct	Reads $5N \leq 2p - 2$ (not $5K$). All five conditions hold throughout $K/3 < p \leq K$; the paper asserts this on p. 11.
(4.10) rank bound	correct	$\text{rank } L = r_p$ <i>exactly</i> at all 37 tested primes — the bound is sharp.
(4.11), (4.12)	correct	Weights re-derived; $v_p^G(A_{uv}) \geq w_u + w_v$ with 0 failures in $\approx 3 \times 10^5$ entries.
divided difference (p. 12)	imprecise	Conclusion true, but needs the $-\frac{1}{4} + \frac{1}{2j}$ tail of (2.3): over 2087 node pairs the congruence holds 2087/2087 with the tail and 0/2087 without. One displayed line missing.

continued on the next page

Statement	Verdict	Comment
Lemma 4.2, (4.13)	correct	Proved independently; 6000 random trials, 0 violations, bound attained. The nonpositive <i>half</i> -integer hypothesis is demonstrably necessary (fails for quarter-integers in 102/3000 trials).
(4.14), table p. 13	correct	Rebuilt from (4.11)–(4.12)+Lemma 4.2: 0 mismatches over 3125+14100 (K, p) pairs, and every row of the cost table re-derived.
Proposition 4.3	correct	Verified against exact Δ_K at $K = 40, 80, 120$: <i>0 violations at all 37 primes, equality at 27</i> . Second assertion ($p > K$) exact at every prime up to $4h$.
Section 5		
(5.1)	correct	The three branches partition the primes with no gap or overlap; boundary cases $pM = K$, $3p = K$, $p = 2h$, $p > 2h$ all check. Branch 1 bounds F_K , branches 2–3 bound Δ_K — load-bearing, and correct.
(5.2), (5.3)	correct	Only $p \leq 2h$ enters; (5.3) is exactly Legendre applied to (2.5), signs included.
Proposition 5.1	not checked	Correct bookkeeping of (3.12), 4.1, 4.3. Inherits Proposition 4.1’s status.
(5.4)–(5.6) $\Gamma, \mathcal{N}, R$	correct	Re-derived from (4.4)–(4.8) and (5.3) independently; R affine on exactly 143 intervals of $[3, 20]$, matching the paper’s $s = 143$.
(5.7) uniformity	imprecise	True and better than claimed: $ \gamma_p^{\text{in}} - p\Gamma(K/p) \leq 3.8M$, both signs, uniform in K (exactly 35 at fixed x over a factor 32 in K). The paper’s one-paragraph justification is very terse and gives no constant; “ $O(M)$ intervals” over-counts (there are at most 3).
(5.8), (5.9)	correct	Re-derived from (4.14) as an exact $K \rightarrow \infty$ limit; 0 mismatches in 20000+30000 exact random-rational tests. $\int_{1/3}^{1/2} d = 9/640$ exactly.
(5.10) $I_{\text{out}} = 127751/96000$	correct	Exact, by three independent routes; all 11 rows of Table 4 and their endpoints reproduce.
Proposition 5.2, (5.11)	correct	PNT step correct: $p \leq \sqrt{5K}$ gives $O(K^{3/2} \log K)$; $\lfloor \log_p 5K \rfloor = 1$ exactly on $(\sqrt{5K}, K/M]$; partial summation legitimate. End-to-end control converges to the predicted blocks.
(5.12)–(5.14)	correct	Re-derived by hand; the quadratic coefficient of R cancels <i>identically</i> in (α, λ) once $H = \lambda + 3\alpha$, so convergence of $\int R/x^3$ is structural. $-1/2 \leq Q \leq 13/8$ correct.
(5.15)–(5.17)	correct	Both integrations by parts re-derived; $\overline{P} = 2923/240$, $\max G_0 = 1/(36\sqrt{3})$, $ \mathcal{C} \leq 118511/(4320\sqrt{3}) < 16$. (5.16) is an exact identity of its constants ($= -2689/48000$) and is conservative by 4.05×10^{-3} ; (5.17) holds at $M = 200$ with slack 2.0×10^{-5} .
(5.18)–(5.21)	correct	$\int_3^{20} R/x^3$ exact; A_* , A_{200} , A_{100000} reproduce exactly.
Section 6		
(6.1), (6.3) V , C_*	correct	V is exactly the field produced by $y = K\sqrt{t}$; $C_* = 2.65303599034048866$ is exactly the K^2 coefficient of $\log S_K$ (symbolic difference 0).

continued on the next page

Statement	Verdict	Comment
Lemma (6.2)	6.1, correct	<i>The crux, and it is both true and proved.</i> $\sup_{t \geq 0} (2U^\rho - V) = -6.6498938806$ at $t^* = 0.6195486669$, against $M_0 = -6.645$: slack 4.894×10^{-3} . Independently reproduced here.
Lemma (6.4)	6.1, correct	$\lambda M_0 - I(\rho) + C_* = -1.366995564512461 \leq \bar{U} = -1.3669955$. True, but the margin is 6.45×10^{-8} : $\bar{U}$ is literally the 7-decimal rounding of the left side.
Lemma structure	6.1, correct	$\sum c_j = 37/40$ <i>exactly</i> in integer arithmetic; nesting holds; $\min_j (b_j - a_j) = 0.005085947445 > 1/225$.
Lemma (6.5)	6.2, correct	Standard and correctly proved; the integrability hypothesis is exactly what the circle/arcsine application needs.
(6.6)–(6.9)	correct	Re-derived; arcsine window bound $4\sqrt{2r}/(\pi\sqrt{L})$ correct; $\sum_j 8c_j \sqrt{2}/(\pi\sqrt{b_j - a_j}) = 9.42 \leq 60$. Tested end-to-end on 180 random configurations at $K = 40, 80, 200$.
(6.8)	correct	True and sharp: the coefficient $13/10 = 2\lambda - 1 + 6\alpha$ is exact and $6\alpha^3/t$ is exactly what $6\alpha \log(1 + \alpha^2/t) \leq 6\alpha \cdot \alpha^2/t$ gives (an earlier pass called it $3\times$ too generous; that was a misreading).
(6.10) Andréief	correct	Verified symbolically and exactly for $h = 2, 3, 4$, including the $1/h!$; weight identity checked numerically.
(6.11), (6.12)	correct	8192 valid and generous (the sharp constant is $(2\pi)^4/12 = 129.88$). (6.12) correct, with the $t = 0$ case the binding one.
(6.13)–(6.15)	correct	Exponent $2h(h+6N-K) + 16h$ and prefactor $4096e^{12}$ re-derived; $\int_0^\infty t^{-1/2}(1+\sqrt{t})^5 e^{-\sqrt{t}} dt = 652$ exactly; the $K^2 \log K$ terms cancel exactly ($111/160 - 111/160 = 0$).
Proposition (6.16)	6.3, correct	Holds at $K = 40, \dots, 240$ (legitimate instances, since the claim is for every $K \in 40\mathbb{Z}_{>0}$) with room to spare (the excess $\log F_K - \bar{U}K^2$ peaks at $+276$ at $K = 120$ and is $+27$ at $K = 240$, against an allowance $24K \log K + 200K$ rising from 11541 to 79569: a factor 79 at $K = 40$, 387 at $K = 200$, 2988 at $K = 240$). Note its own bound is vacuous for $K \leq 242$.
Section 7		
(7.1)	correct	A legitimate sum of two upper bounds; $Q_{K,M}(\zeta(5)) > 0$ so the logarithm exists.
(7.2)	correct	Both exact margins reproduce digit for digit from Appendix B.3.
Appendix A		
Table 1	correct	Transcribed from the text layer and checksummed: $\sum c_j = 37/40$ exactly.
(A.1), (A.2)	correct	(A.1) against quadrature to $< 10^{-40}$; (A.2) re-derived from nesting and confirmed by double quadrature.
(A.3), (A.4)	correct	Truncation orders ample: remainders 4.9×10^{-64} and 2.1×10^{-51} , far below the claimed 2^{-144} and 10^{37} times below the tightest margin in play.
(A.5), min of V	correct	Matches direct integration of (6.1); Φ'' sign change at $y^2 = 711/880$ exactly; minimum inside (q_-, q_+) as claimed.
(A.6)	imprecise	$V_* = 0.913559526 \leq \min V = 0.913559534$: valid. “The parenthesis is negative” holds by 2.5×10^{-8} but follows in one line from the preceding sentence; terse, not unjustified.

continued on the next page

Statement	Verdict	Comment
(A.7), (A.8), Table 2	correct	The k -column lists whitespace-separated runs. Under that reading the 91 rows give 684 intervals that tile $[0, 2]$ exactly (verified here in exact rational arithmetic). The caption should say so.
(A.9)	correct	$\max \mathcal{B}(l, r) = -6.645002689215588$ over all 684 intervals, below $-6645002/10^6$. Margin 6.892×10^{-7} ; exactly one interval is within 10^{-5} of the threshold.
(A.10), (A.11)	correct	Both enclosures correct; $\kappa = 13.8658566788652898$ inside (A.11) and $< 6933/500$.
Appendix B		
(B.1)	correct	Re-derived from $\ell(x, z) = \lfloor x - z \rfloor + \lfloor x + z \rfloor + 1$.
(B.2) break-points	imprecise	Correct and complete (142 interior points, $s = 143$), but redundant: $2(1 - \alpha) = 2\lambda$ is listed twice and the 2α -points are a subset of the 4α -points.
(B.3), Tables 3, 4	correct	All 17 rows of Table 3 and all 11 rows of Table 4 reproduce exactly; the 17 rows sum <i>exactly</i> to (5.18). (See Section 7.2 on a disputed digit.)
B.3 final margins	correct	Both printed rationals reproduce exactly as fractions.
Appendix C		
Proposition C.1, (C.1)	correct	(C.2)–(C.5) and the combination re-derived; $\kappa < 6933/500$ with margin 1.433×10^{-4} . The full chain was run numerically at $K = 40$ and 80 : (C.3) holds as an exact identity to 401 and 1500 digits.
(C.4)	correct	Uses $m! \geq (m/e)^m$ and monotonicity of $x \log(eK/(\alpha x))$; valid since $h = 37K/40 < 40K/3$.
(C.5)	imprecise	The displayed $h^2 \mathcal{R}^{2h}$ is reproducible from the paper’s own argument and is true; it is weaker by the $O(1)$ factor $\mathcal{R}^2 = 293.8$ than the $h^2 \mathcal{R}^{2(h-1)}$ the same step gives. κ unaffected.
(C.6)	correct	$\kappa = 13.86585667886528983$ by two evaluators agreeing to > 100 digits; margin to ϱ is 1.4332×10^{-4} . The “positive margin absorbs the polynomial factor” only for $K \gtrsim 2.81 \times 10^5$; neither number is given.
(C.7), C.2	correct	All rational identities exact: $\varrho c - 260 = -1/80$, $\epsilon - \lambda/c = 1/24000$. The perturbation argument genuinely excludes a zero. Depends on Proposition 5.1 as well.
(C.8)	correct	$\lambda \varrho - \epsilon = 511067/40000 = 12.776675 < 13$ exactly.
C.3 primitive normalization	correct	Gauss’s lemma; and it is the remark that makes the theorem robust to an error in §§3–5 (see Section 5).

3 Per-section commentary

3.1 Sections 1 and 2: the construction

Nothing here is wrong, and the parts a referee would most want to see are the parts that are most solid.

The final logical step is airtight and deserves to be stated plainly, because it is what makes the paper’s strategy different from a linear-form construction: one needs only $Q_n \in \mathbb{Z}[X]$,

$\deg Q_n \leq 37n$, $Q_n(\zeta(5)) > 0$ and $Q_n(\zeta(5)) < e^{-139n^2/5}$. Then $b^{37n}Q_n(a/b)$ is a positive integer smaller than $\exp(37n \log b - \frac{139}{5}n^2) \rightarrow 0$. The decay is quadratic in n and the height cost is linear, so *no estimate on the coefficients of Q_n is required at all*. This is why the paper can afford to work with polynomials of growing degree.

The functional of (2.2)–(2.3) is well defined and there is no circularity, contrary to what a quick reading might suggest. The set $\{t^e\}_{e \geq 0} \cup \{1/(t+j^2)\}_{j \geq 1}$ is a basis of the space of rational functions with simple poles in $\{-j^2\}$, so (2.2) and (2.3) prescribe values on a basis and determine a unique $\mathbb{Q}$ -linear functional; the word “extend” in the paper is loose for “define”. One hypothesis is used and never stated: $0 = -0^2$ is not an allowed pole (all products run over $j \geq 1$), which is exactly why the integral in Proposition 2.2 converges.

Equation (2.9) is correct including the sign, and this matters: it is what gives $\deg Q_{K,M} = h$ *exactly*, which the final step needs. We verified it symbolically on small cases in both sign parities, and then on the paper’s own smallest real parameters $(N, K) = (3, 40)$, where $\det[X]G_{40}$ equals $(-1)^{666} \prod_{j=4}^{40} j^4 D_3(-j^2)^5$ exactly, as a 1594-digit integer, reproduced by two tracks with different determinant routines. A structural fact the paper never states is worth knowing: $[X]\mu_X(D_N^6 t^m / D_K)$ is an $(h-1)$ st divided difference and vanishes identically for $m < 22n - 3$, so $[X]G_K$ carries a large zero block and the Vandermonde nonvanishing in (2.9) is genuinely load-bearing rather than decorative.

Proposition 2.2 was checked step by step and is right, including both correction terms and both signs in (2.3) — the place where such a computation usually goes wrong. The four integrations by parts were certified by comparing two different integrands to 45–51 digits; Hermite’s formula (DLMF 25.11.29) was matched to ≥ 24 digits at five values of a . Positive definiteness of $G_K(\zeta(5))$ was confirmed at $K = 40, 80, 120, 160, 200, 240$, and at $K = 40$ certified five ways including interval arithmetic. A warning for anyone reproducing this: the entries carry up to 107 digits of cancellation, and evaluating Δ_{40} at $\zeta(5)$ needs ≥ 2600 digits — at 400 digits it returns +1203.8, pure noise. The single omission in the proof is the interchange of $\sum_{\ell \geq 1}$ with $\int_0^\infty$, which is Tonelli in one line since every term is positive.

3.2 Section 3: the local functional

No error, and — the thing we were most concerned to check — no place where a power of p is gained or lost silently. Some 400 automated checks in exact arithmetic with explicit valuations, plus a SageMath/ $\mathbb{Q}_p$ cross-check of the Bernoulli inputs, produced no failure.

The decisive test is (3.11). A change of basis is exactly where a construction of this kind loses a power of p without anyone noticing, and here the change of basis to the integer-valued f_i is an *exact identity*: with $q_i = \sum_k C_{ik} t^k$ triangular with $C_{ii} = (-1)^i 2/(2i)!$, one gets $\det[(K!)^2 \mu_X(f_i f_j / D_K)] = ((K!)^2 / (N!)^{12})^h (\det C)^2 \Delta_K = S_K \Delta_K = F_K$, with S_K *precisely* the constant printed in (2.5). We verified this exactly for five (K, N, h) triples and then proved it. Nothing is gained or lost by the change of basis, and nothing by the pullback $t = -x^2$.

Lemma 3.1’s hypotheses were attacked individually. $d(r_\nu) < p$ is *necessary*: at $r = p$ the conclusion fails with $v_p = -5$. $\deg U_0 \leq p + 1$ is sufficient but far from tight — the true threshold is $4p - 2$, because $\kappa_d = d(d-1)(d-2)B_{d-3}/24$ has a compensating factor of p at $d = p + 2, 2p + 1, 3p$, and the first loss is at $d = 4p - 1$ (measured: 27, 43, 51, 67, 75, 91 for $p = 7, 11, 13, 17, 19, 23$). The third hypothesis, $r_\nu - r_\eta \in \mathbb{Z}_p^\times$, is *redundant* given $d(r_\nu) < p$: inside $\{-p, \dots, p-1\}$ a congruent pair must be $\{r, r-p\}$, and $H_{p-1-r}^{(5)} \equiv H_r^{(5)} \pmod{p}$ makes

the two p^{-1} -sized residue contributions cancel. This is unused slack, not an error.

Lemma 3.2 is correct and the exponent p^{-4} is right (p^{-1} from Raabe’s multiplication theorem, p^{-3} from three differentiations). $Y = p^5 X + C_p$ is exactly the choice making the right-hand side at $g = 1/x$ equal $-X$. The hypothesis $p \geq 7$ is genuinely required, not decorative: $v_p(\sum_{a < p} a^{-4}) = 1$ for $p \geq 7$ but 0 for $p = 5$. One refinement for the record: $C_p \in p^5 \mathbb{Z}_p$ for every $p \geq 7$ is a theorem, but $v_p(C_p) = 5$ *exactly* only when $p \nmid B_{p-5}$; at the irregular prime $p = 37$ one has $v_{37}(C_{37}) = 6$. The paper claims only membership, so it is unaffected.

The one defect in §3 is cosmetic: in Lemma 3.3’s proof the residue coefficient is written $c_r = \pm(K!)^2 r A(r) / ((K+r)!(K-r)!)$ with the sign left to the reader. It is uniformly $(-1)^{K-r}$, and only $|c_r|_p$ is used, so nothing depends on it.

3.3 Section 4: the local determinant estimates

This is where the risk lives, and it is worth being precise about *which* risk.

The outer range is verified, and it is essentially sharp. We computed $\Delta_K(X)$ exactly in $\mathbb{Q}[X]$ at $K = 40, 80, 120$ (coefficients up to 2.7×10^4 digits) by two independent implementations, and compared $v_p^G(\Delta_K)$ with γ_p^{out} at every prime satisfying (4.9). There are **0 violations at all 37 primes, with equality at 27 of them**. A formula of this complexity that is exactly attained at 27 independent primes is almost certainly right. The second assertion of Proposition 4.3 ($v_p^G(\Delta_K) \geq 0$ for $p > K$) is exact at every prime up to $4h$, which is nontrivial because the residue denominators contain primes up to $2K - 1$ and they cancel. Every ingredient was separately verified: unimodularity of the basis (4.11) at 37/37 primes; rank $L = r_p$ *exactly* at 37/37, so the rank bound (4.10) is sharp; $v_p^G(A_{uv}) \geq w_u + w_v$ with 0 failures in $\approx 3 \times 10^5$ entries; and the combinatorial identity (4.14) rebuilt from (4.12) and Lemma 4.2 with 0 mismatches over more than 1.7×10^4 (K, p) pairs. Where γ_p^{out} is not sharp — at $K \geq 2p$, where it misses by 78, 60, 17 at $K = 120$ — the loss is in the *safe* direction: it inflates A_M and makes (7.2) harder. Section 4 therefore cannot rescue a false theorem, which is a useful thing to know: if the paper is wrong, it is not wrong here.

Lemma 4.2 we proved independently and stress-tested (6000 random instances over $\mathbb{Z}_7, \mathbb{Z}_{11}, \mathbb{Z}_3$; 0 violations; the bound attained in every sharpness test). Its restriction to nonpositive half-integer weights is not a convenience: with quarter-integer weights the conclusion fails in 102 of 3000 random instances, e.g. $p = 11, h = 3, w = (-3/2, -1/4, -3/2), r = 1, z = 0$, where the claimed bound is $-13/2$ and the true valuation is -7 . The paper says the restriction is essential and it is right.

The inner range: the proof was read and reconstructed; only the numerical cross-check is missing. Proposition 4.1 is proved in the paper on pp. 10–11 — about twenty-five lines, resting on Lemmas 3.1 and 3.2 and on the allocation (4.4)–(4.8), with the side conditions displayed in the text. We reconstructed that argument line by line, twice and independently, and found it complete apart from the one asserted inequality of Finding 1, which we prove below. We record two things about our confidence in it. First, it is a reading, not a verification: it is the one substantial argument in the paper whose correctness rests on our having followed it rather than on a machine check. Second, and unlike every other estimate here, its conclusion cannot be compared against a computed determinant: hypothesis (4.1) forces $p > 200M \geq 8000$ and $K \geq 200M^2 \geq 320000$, hence $h \geq 296000$, and at the computable $K \leq 160$ the allocation (4.4) is degenerate ($L_0 = 4M + 10 > h$, and $T < 0$). So the corroboration we can give it is indirect:

- *The mechanism, at the smallest nondegenerate instance.* At $K = 80, p = 23$ with L_0

freed, the true value is $v_{23}^G(\Delta_{80}) = -188$. With $L_0 = 3$ or 4 (all side conditions satisfied) the entry inequality holds with **0 failures** and $\gamma_p^{\text{in}} = -188$ *exactly*. With $L_0 = 0, 1, 2$ (zero-source dominance fails) the entry inequality fails 2094/812/29 times and γ_p^{in} would be $-170/-180/-186$, i.e. a *false* bound. The test is therefore discriminating, and the paper’s side conditions are exactly the right ones.

- *The side conditions, at the real parameters.* All of them — $L_a \geq 0$, the two Lemma-3.1 degree restrictions, zero-source dominance, the dimension identity $L_0 + \sum_a L_a = h$, and $2Hx - 21/20 < T < 2Hx$ — were swept over all 9158 inner primes at $M = 40$, $K = 200M^2$, and over 27171+13960 sampled primes at $M = 200$, $K = 8 \times 10^6$ and 4×10^7 . **No failures**, with three to four orders of magnitude of slack in the degree conditions.
- *The hand-off from §3.* The sharpest worry raised in the audit was whether §4 actually produces an honest *polynomial* U_0 of degree $\leq p + 1$ for Lemma 3.1, given that the raw pullback numerator has degree $184n$. It does, and the paper displays exactly this on p. 11: $\nu_i(c) + \nu_j(c) + 6\ell_N(c) \leq 2(T + 1) < \frac{23}{5}M + 2$ at an ordinary source and $5 + 4L_0 + 12m_N \leq \frac{169}{10}M + 45$ at the zero source, both $\leq p + 1$ since $p > 200M$. We verified both over the same sweeps. This closes the hand-off.

What remains without an independent machine check is the p -adic determinant inequality $v_p^G(\Delta_K) \geq \gamma_p^{\text{in}}$ itself at admissible parameters, and with it the $m_N \geq 1$ (i.e. $p \leq N$) part of the inner range, which for $M = 200$ is most of it. To be clear about what that does and does not mean: the paper proves this inequality, and we believe the proof; what we cannot do is confirm it the way we confirmed Proposition 4.3, by computing the determinant and comparing. Certification here needs a referee or a formalisation, and §5 says so in those terms.

Finding 1. (p. 10, the line after (4.4); the only genuine gap in the written mathematics.) The paper writes “From (4.4), $2Hx - \frac{21}{20} < T < 2Hx$, $b_a \leq 6\alpha x + 3$.” The attribution is wrong — $b_a \leq 6\alpha x + 3$ is a fact about ℓ_N , not a consequence of (4.4) — and no proof is given.

Why it matters. The naive estimate $\ell_N(a) \leq 2\lfloor N/p \rfloor + 2$ gives only $b_a \leq 6\alpha x + 6$, under which the paper’s own chain reads $T - b_a > 2\lambda x - \frac{141}{20}$, which at $x = 3$ equals $-3/2$: the nonnegativity of the dimensions L_a fails and Proposition 4.1 collapses. *Why it is true.* Writing $\ell_N(a) = 2m_N + \mathbf{1}_{a \leq v_N} + \mathbf{1}_{a \geq p - v_N}$ with $v_N = N \bmod p$, both indicators can be 1 only if $v_N \geq (p + 1)/2$, and in that case $\alpha x - m_N = v_N/p > 1/2$; so $\ell_N(a) < 2\alpha x + 1$ either way. *Evidence.* We checked the inequality over all 9158 inner primes at $M = 40$, $K = 320000$ (and 17214 at $K = 640000$, 190197 at $M = 200$, $K = 8 \times 10^6$): no violation, and the bound is *strict everywhere* but only just — the slack is $6v_N/p$ when $1 \leq v_N \leq (p - 1)/2$ and $6v_N/p - 3$ when $v_N \geq (p + 1)/2$, so it becomes arbitrarily small as v_N/p approaches 0 or $1/2$. The observed minimum is 0.00119, at $p = 47981$, $v_N = 24000$ (an earlier track recorded “equality whenever $p \mid N$ ”; that is wrong — at $p \mid N$ the slack is exactly 3). With the paper’s bound the chain gives $T - b_a > 2\lambda \cdot 3 - \frac{81}{20} = +3/2$ exactly, so there is no room to spare at $x = 3$ — this is not a place to be terse. The author should be asked to insert the case split.

3.4 Section 5: normalization and the prime sum

Section 5 and Appendix B are *correct*, conditional on (3.12), (4.8) and (4.14). Every item reproduced exactly in independent implementations. The case analysis of (5.1) is exhaustive and non-overlapping, and one subtlety is worth recording because it is load-bearing and right:

branch 1 bounds $v_p^G(F_K)$ directly via (3.12) and therefore correctly *omits* $v_p(S_K)$ from L_p , whereas branches 2–3 bound $v_p^G(\Delta_K)$ and must include it. Since $v_p(S_K) \sim -1110n^2/(p-1) < 0$ for small p , branch 1 would be false if (3.12) were a statement about Δ_K rather than F_K . It is about F_K (p. 9).

The single weak point of the exposition is (5.7). The claim that $\gamma_p^{\text{in}} = p\Gamma(K/p) + O_M(1)$ uniformly in the inner range carries the entire arithmetic constant and is justified in one informal paragraph with no explicit constant and no displayed estimate. The claim is *true*, and in fact better than the paper suggests: computing γ_p^{in} exactly from (4.4)–(4.8) over the full inner range gives $|\gamma_p^{\text{in}} - p\Gamma(K/p)| \leq 3.8M$, taking both signs, uniform in K — at fixed $x = K/p \approx 10$ the difference is *exactly* 35 for K ranging over a factor of 32. So the error is $O(M)$, not $O(M^2)$. The mechanism is a cancellation rather than an omission: the reserved zero block (4.7) is genuinely $\Theta(M^2)$ (it contributes +13734 at $p = 8081$, $K = 320000$) but is almost exactly cancelled by the ordinary block because $L_0 = 4M + 10$ enters (4.4) and lowers T . That is precisely what the paper’s sentence about “altering the number of allocated rows by $O_M(1)$ ” means, but a reader cannot see it. A minor slip in the same paragraph: “There are $O(M)$ such intervals” over-counts — in the z variable the integrand of (5.4) is a step function with at most three pieces. The over-count is in the safe direction.

Everything else in §5 is exact. R is affine on exactly 143 intervals of $[3, 20]$, matching the paper’s $s = 143$, with zero affineness failures on any of them; $\int_3^{20} R/x^3$, $I_{\text{out}} = 127751/96000$, $\int_{1/3}^{1/2} d = 9/640$, A_* , A_{200} and A_{100000} all reproduce as exact rationals by more than one route. The tail bounds (5.16) and (5.17) are valid and conservative: the true $\int_{20}^{\infty} R/x^3 = -0.06007376$ against the printed $-2689/48000 = -0.05602083$, so the paper discards 4.05×10^{-3} there, which is 23% of its own final margin.

3.5 Section 6: the real bound

Correct, and — this is the part of the audit that changed most — *proved*, not merely true. Our initial reading recorded Lemma 6.1 as “could not check” because Appendix A’s certification looked unverifiable; it is verifiable, and it verifies. See Section 3.7.

Lemma 6.2 is a correct proof of a standard fact, with the integrability hypothesis exactly matched to the circle-plus-arcsine application. It is applied correctly: $\sigma = K^{-1} \sum \omega_i$ and ρ are both compactly supported and $\sigma - \rho$ has mass $h/K - \lambda = 37n/40n - 37/40 = 0$ exactly. The regularization bookkeeping in (6.6)–(6.9) is right in every direction, and the constant 60 is honest (the true value of $\sum_j 8c_j \sqrt{2}/(\pi \sqrt{b_j - a_j})$ is 9.42; even the crude route using only $b_j - a_j > 1/225$ gives 49.97). The hypothesis $b_j - a_j > 1/225$ is used in exactly one place, here.

The scaling deserves a note for anyone reading the paper: it is $y_i = K\sqrt{t_i}$, i.e. $y^2 = K^2 t$, and only that scaling makes the bookkeeping in (6.13) close. We re-derived that exponent tracking every power of K — Vandermonde $K^{2h(h-1)}$, and per variable $K^{12N} \cdot K^{-2K} \cdot K^{12} \cdot K^5 \cdot K^1$ — and obtained $2h(h + 6N - K) + 16h$, exactly as printed, with prefactor $8192/2 \cdot e^{12} = 4096e^{12}$, also as printed. The integral $\int_0^{\infty} t^{-1/2}(1 + \sqrt{t})^5 e^{-\sqrt{t}} dt = 652$ is exact ($= 2e \Gamma(6, 1)$). The $K^2 \log K$ terms of (6.14) and (6.15) cancel exactly, and the K^2 coefficient of (6.15) equals C_* of (6.3) with symbolic difference zero. Andréief’s identity (6.10) is exact for $h = 2, 3, 4$ including the $1/h!$.

How much room §6 actually has. Four tracks independently solved the underlying weighted equilibrium problem — by a constrained convex QP with exact cell energies, by one-cut resolvent inversion, by a Fourier–cosine Frostman solve, and by Frank–Wolfe on a log-spaced grid, each with controls reproducing the semicircle, Marchenko–Pastur and quartic-field laws —

and they agree to ten digits: the true optimum is $E^* = \sup_{\mu}(I(\mu) - \int V d\mu) = -4.0385457735$, on $\text{supp} = [7.12 \times 10^{-5}, 0.753950]$ with Robin constant -6.6607811230 . The paper’s comparison bound $\lambda M_0 - I(\rho) = -4.0200316$ is *above* E^* , hence valid, and loose by 0.01851 . In other words the true constant is $\bar{U}_{\text{true}} = E^* + C_* = -1.3855098$ against the paper’s $\bar{U} = -1.3669955$: **Section 6 holds 0.0185 of unused reserve, slightly more than the entire final margin** $|A_{200} + \bar{U}| = 0.0174$. The real side is conservative, not optimistic. This is confirmed from the other direction by direct measurement: $K^{-2} \log F_K(\zeta(5))$ at $K = 40, \dots, 200$ extrapolates to $-1.3854\dots - 1.3869$, and a model fitted on $K \leq 160$ predicts the $K = 200$ value with an out-of-sample error of 1.4×10^{-6} .

The consequence for the audit is sharp: *any failure of this paper must be located in Sections 3–5, not in Section 6*. If the arithmetic side were off by as much as $+0.036$, the proof could in principle still be rescued by replacing ρ with the true equilibrium measure.

3.6 Section 7

Two lines, both correct. (7.1) is a legitimate sum of two upper bounds, and $Q_{K,M}(\zeta(5)) > 0$ so the logarithm exists. (7.2) reproduces digit for digit. The only observation is that (7.1) is a limsup statement while (2.7)–(2.8) are “for all sufficiently large n ”: no threshold anywhere in the paper is effective. That is legitimate for an irrationality proof, but it means no finite computation can ever confirm or refute (2.7) directly — every check must be of the asymptotic constants, which is what this audit did.

3.7 Appendix A

Appendix A proves Lemma 6.1, and it does prove it. Because the audit initially doubted this, we state the reproduction in full.

Table 1 gives $10^{12}a_j$, $10^{12}b_j$, $10^{12}c_j$ as integers. Taken from the PDF text layer (*not* from the rendered page image — see Section 7.2) it checksums: $\sum c_j = 37/40$ *exactly* in integer arithmetic, the nesting $0 < a_{16} < \dots < a_1 < b_1 < \dots < b_{16} < 2$ holds, and $\min_j(b_j - a_j) = 0.005085947445 > 1/225$, attained at $j = 1$. From (A.2), $I(\rho) = -2.126593445147050403254$, inside the printed enclosure (A.10); and $C_* = 2.653035990340488661129$, also inside. Hence

$$\lambda M_0 - I(\rho) + C_* = -1.366995564512460935617 \leq \bar{U} = -\frac{2733991}{2000000},$$

which is (6.4) — **with a margin of 6.45×10^{-8}** . $\bar{U}$ is literally that number rounded up at the seventh decimal.

The pointwise inequality (6.2) holds comfortably: $\sup_{t \geq 0}(2U^\rho(t) - V(t)) = -6.6498938806$ at $t^* = 0.6195486669$, against $M_0 = -6.645$, a slack of 4.894×10^{-3} ; on $[2, 10^8]$ the supremum is -8.27 . We reproduced the field V of (A.5) against direct quadrature of (6.1) to 5×10^{-51} as a control.

The certification itself, (A.7)–(A.9), is the part that looked unverifiable, because Table 2’s k -column contains rows like “19 3 0 2,3” and “21 5 2, ..., 4 10, ..., 13” whose format the caption does not explain. The rule is that the column lists a *union of whitespace-separated runs*. Under that reading all 91 rows parse, and the resulting **684 intervals tile $[0, 2]$ exactly** — we checked this in exact rational arithmetic, gap-free and overlap-free within each of the 35 blocks $[A_j, A_{j+1}]$. Evaluating $\mathcal{B}(l, r)$ of (A.7) on all 684 gives

$$\max_{(j,d,k)} \mathcal{B}(l, r) = -6.645002689215588, \quad \text{at } (j, d, k) = (30, 6, 26),$$

$$[l, r] = [0.5480159423, 0.5492641749],$$

which is below the printed $-6645002/10^6$. **So (A.9) is true, with a margin of 6.892×10^{-7} ,** and exactly one of the 684 intervals lies within 10^{-5} of the threshold. The prescribed truncation orders are ample: (A.3) at $m = 64$ on $z \leq 1/3$ has remainder 4.9×10^{-64} and (A.4) at $m = 80$ on $z \leq 1/2$ has remainder 2.1×10^{-51} , both far below the claimed $2^{-144} \approx 4.5 \times 10^{-44}$ and some 10^{37} below the tightest margin in play. A margin of 6.9×10^{-7} on quantities of size 6.6 is resolved by double precision, let alone by the 25–50 digits we used; the whole appendix verifies from the printed tables in seconds (our own run: 4 s; the slowest of the four independent reproductions took two minutes).

Two remarks. First, the author is operating at the edge of his own stated precision in three places — (6.4) at 6.45×10^{-8} , (A.9) at 6.89×10^{-7} and (5.17) at 2.0×10^{-5} — and one fewer dyadic refinement on the worst Table-2 cell would break (A.9) as written. Second, and more seriously as a matter of robustness: the slack in (6.4) is the same order as a single transposed digit pair inside one Table 1 entry. We confirmed this experimentally: reading $10^{12}b_8$ as 201105672729 instead of the correct 201105762729 moves $I(\rho)$ by 3.3×10^{-8} . The printed table is self-consistent and correct, but a referee is entitled to ask that (6.2) be quoted at -6.6498 (slack 4.9×10^{-3}) rather than at $M_0 = -6.645$. That single change replaces $\bar{U}$ by -1.3715224 and turns the margin in (7.2) from 0.052484 into 7.29553 — a factor of 139 — at no cost anywhere else. (We recompute this here; the figure “280” that circulated in the audit’s drafts was wrong.)

3.8 Appendix B

Exact throughout. All 17 rows of Table 3 and all 11 rows of Table 4 reproduce, the 11 row endpoints coincide with independently derived breakpoints, and the 17 rows of Table 3 sum *exactly* to (5.18). (B.1) re-derives from $\ell(x, z) = \lfloor x - z \rfloor + \lfloor x + z \rfloor + 1$. The quadratic-cancellation identity $2\lambda - 12\lambda\alpha + 2(H^2 - H - \lambda^2) - 18\alpha^2 + 6\alpha = 0$ holds exactly, and in fact vanishes *identically* in (α, λ) once one notes $H = \lambda + 3\alpha$ (which (4.4) forces but the paper never says): the convergence of $\int R/x^3$ is structural, not a numerical accident of $3/40$ and $37/40$. Stating this would make §5.3 considerably more convincing. The breakpoint set (B.2) is correct and complete but redundant: $2(1 - \alpha) = 2\lambda = 37/20$ is listed twice, and the 2α -points are a proper subset of the 4α -points.

3.9 Appendix C

Correct. Proposition C.1’s chain (C.2)–(C.5) was re-derived and then run numerically end to end at $K = 40$ and $K = 80$, where (C.3) holds as an exact identity to 401 and 1500 digits — this confirms the residue structure, the rescaling and the exponent $12N - 2K + 6$ simultaneously. The constant is $\kappa = 13.86585667886528983095$ by two evaluators agreeing to more than 100 digits, against $\varrho = 6933/500 = 13.866$: (C.6) is true with margin 1.4332×10^{-4} , and the paper’s own enclosure (A.11) is correct. All the rational identities are exact: $\varrho c - 260 = -1/80$, $\epsilon - \lambda/c = 1/24000$, $\lambda\varrho - \epsilon = 511067/40000 = 12.776675 < 13$. The perturbation argument in C.2 genuinely excludes a zero, and the exponent 260 is essentially forced by this method (its best possible value is 259.54).

Three caveats the paper should state. (i) “Its positive margin absorbs the polynomial factor in h ” needs $K \gtrsim 2.81 \times 10^5$; neither the margin nor the threshold is given. (ii) The true norm is much smaller than the bound: $\log \|B_K\|/K = 7.314$ at $K = 40$ and 7.388 at $K = 80$ against the asserted $\varrho = 13.866$, so Proposition C.1 has roughly a factor-2 margin in the exponent

wherever it is testable, and the exponent 260 is an artefact of the estimate — a proved rate near 7.4 would give about 139 by the identical argument. (iii) Corollary 1.2’s threshold is not merely unspecified but *ineffective*: $M = 100000$ forces $K \geq 200M^2 = 2 \times 10^{12}$, hence $\log b \geq 1.067 \times 10^{11}$, i.e. $b_0 > \exp(1.07 \times 10^{11})$. A larger K is no escape, since e^{e^K} then blows up. Finally, Corollary 1.2 depends on Proposition 5.1 and on Theorem 2.1’s degree claim as well as on (7.1), Proposition 2.2, §2.3 and (C.6) — without integrality, $b^h Q_K(a/b)$ is not an integer and the contradiction evaporates.

4 The decisive numerics

4.1 The two constants and the balance

Everything in Table 2 was recomputed here in exact rational arithmetic from the paper’s own formulas and matches the printed values.

Quantity	Value	Status
I_{out}	$127751/96000 = 1.3307395833$	exact, three routes
$\int_3^{20} R x^{-3} dx$	0.0427881769084	exact, matches (5.18)
A_*	1.3175069269083709	exact, matches (5.19)
A_{200}	1.34958769774170422	exact, matches B.3
A_{100000}	1.31757167571548622	exact, matches B.3
$\bar{U}$	$-2733991/2000000 = -1.3669955$	paper’s constant
$\bar{U}_{\text{true}}$	-1.3855098	true limit; $\bar{U}$ valid, loose by 0.0185
$A_{200} + \bar{U}$	-0.017407802258296	< 0 : irrationality holds
relative margin	1.2899% of A_{200}	the margin the theorem turns on
vs. $-139/8000$	slack 3.2802×10^{-5}	0.0024%: the printed exponent is saturated
$A_{100000} + \bar{U}$	-0.049423824284514	$< -79/1600$, slack 4.882×10^{-5}
$A_{200} + \bar{U}_{\text{true}}$	-0.0359221	the margin with the sharp real constant

Table 2: The balance. Both B.3 margins reproduce exactly: $-1600(A_{200} + \bar{U}) - \frac{139}{5} = \frac{3089837638249482469}{58872425992515135000}$ and $-1600(A_{100000} + \bar{U}) - \frac{7907}{100} = \frac{29873543950273155160680943}{3679526624532195937500000000}$.

Two sensitivities are worth separating, because they have different consequences. *Theorem 1.1* needs only $A_{200} + \bar{U} < 0$ and has 1.29% of room (and 1.60% if one uses (5.11) directly instead of the deliberately lossy (5.20)). The *printed constants* $139/5$ and $7907/100$ survive perturbations of only 3.28×10^{-5} and 5.07×10^{-6} ; any correction above those forces them to be reprinted and breaks Corollary 1.2, while leaving Theorem 1.1 standing. Note also that I_{out} alone is 98.60% of A_{200} : the arithmetic constant is governed by (4.14) and Table 4 — that is, by whether Proposition 4.3 and Lemma 4.2 are true — and not by any arithmetic Section 5 performs.

4.2 The direct computations

$\Delta_K(X)$ was built exactly in $\mathbb{Q}[X]$ from (2.1)–(2.6) alone, by four independent implementations (multimodular determinant plus Lagrange interpolation; rational matrix inverse plus characteristic polynomial; entry assembly from the closed residue formula plus Lagrange interpolation over $\mathbb{Q}$; Berkowitz over $\mathbb{Q}[X]$), with controls passing first (Hilbert determinants through the

same code path; (2.2) against the moment integral; (2.3) against $\int w/(y^2 + j^2)$). All four agree coefficient by coefficient.

K	$\log F_K(\zeta(5))$	$K^{-2} \log F_K(\zeta(5))$	$\log(1/\text{cont})$	$K^{-2} \log P_K(\zeta(5))$
40	-2041.19514254	-1.2757470	1776.06607557	-0.1657057
80	-8508.29536397	-1.3294212	7674.96382916	-0.1302081
120	-19408.86914398	-1.3478381	17604.86199438	-0.1252783
160	-34743.05318075	-1.3571505		
200	-54510.86712322	-1.3627717		
240	-78712.313	-1.36653		

Table 3: Exact determinant data. $G_K(\zeta(5))$ is positive definite at all six K (Proposition 2.2), and (6.16) holds at all six. $P_K = \Delta_K / \text{cont}(\Delta_K)$ is the primitive polynomial, so S_K cancels identically from $K^{-2} \log P_K(\zeta(5))$, and the fourth column is $\log(1/\text{cont}(F_K))$, i.e. the true minimal normalizer. $K \leq 120$ was produced by four independent implementations and $K \leq 200$ by two, each at two precisions; the $K = 240$ row comes from a single track at 400–800 digits and is quoted to its stated precision.

The last column of Table 3 is, in our view, **the single most informative number in the audit**. It is the decay actually achieved by the best possible normalization, and it must be compared with the requirement $< -139/8000 = -0.0173750$. The measured values $-0.166, -0.130, -0.125$ extrapolate to roughly -0.12 (plausible range $[-0.27, -0.072]$), and the value predicted from the verified asymptotics is $A_\infty + \bar{U}_{\text{true}} = 1.3134540 - 1.3855098 = -0.0720558$. *The construction has about seven times the room it needs*. For calibration, the constructions in this circle of ideas that are known to fail do so by a positive margin: the Brown–Zudilin $\zeta(5)$ cellular forms give $+0.22$ per $N^2/2$ with true denominators, and Rivoal’s $(3; 1^q)$ families at $q = 11$ miss by $+0.05$ to $+0.14$. This construction is not marginal in the way those are; only the paper’s *printed exponent* is marginal.

4.3 The local bounds against the truth

K	$v_p^G(\Delta_K) - \gamma_p^{\text{out}}$ at the primes satisfying (4.9), in increasing order of p
40	1, 1, 0, 0, 0, 0 ($p = 17, \dots, 37$)
80	39, 15, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0 ($p = 29, \dots, 79$)
120	78, 60, 17, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 ($p = 41, \dots, 113$)

Table 4: Proposition 4.3 against exact determinants: 0 violations at 37 primes, equality at 27. The slack that does occur is confined to $K \geq 2p$ and is in the safe direction.

For the inner range the corresponding test is impossible, and Table 5 records what was done instead.

Finally, the limiting functions really are the limits: $\gamma_p^{\text{in}} - p\Gamma(K/p)$ is bounded by $3.8M$ uniformly in K and is *exactly* 35 at fixed $x \approx 10$, $M = 40$, for $K = 3.2 \times 10^5, \dots, 1.024 \times 10^7$; $v_p(S_K) - p\mathcal{N}(K/p)$ is likewise K -independent at fixed x ; and $-\gamma_p^{\text{out}} - K(R_0 - d)(p/K)$ has modulus at most 7. Hence A_M is the right arithmetic constant for the local bounds the paper proves.

L_0	side conditions	entry-bound failures	γ_p^{in} (truth: -188)
0	zero-source dominance fails	2094	-170 (<i>false bound</i>)
1	fails	812	-180 (<i>false bound</i>)
2	fails	29	-186 (<i>false bound</i>)
3	all hold	0	-188 (exact)
4	all hold	0	-188 (exact)

Table 5: The mechanism of Proposition 4.1 at the smallest nondegenerate instance, $K = 80$, $p = 23$, where $v_{23}^G(\Delta_{80}) = -188$. The paper’s side conditions are exactly what separates a true bound from a false one.

5 What would have to be true for the proof to work

Stripped of everything that has been verified, the paper rests on the following load-bearing statements. This is the list the author should be asked to defend.

1. $v_p^G(\Delta_K) \geq \gamma_p^{\text{in}}$ **for the inner primes** $K/M < p \leq K/3$ (**Proposition 4.1**). The one substantive claim no computation in this audit could reach. Everything around it checks out — the combinatorics at the real parameters, the mechanism at $K = 80$, $p = 23$, the hand-off degrees, the limit Γ — but the inequality itself is unverified at admissible parameters, and if γ_p^{in} over-claims, then $Q_{K,M} \notin \mathbb{Z}[X]$ and both Theorem 1.1 and Corollary 1.2 collapse silently.
2. **Lemmas 3.1 and 3.2, at the parameters §4 uses them.** A one-power-of- p error would move γ_p^{in} by about h and change A_M . Both lemmas were verified exactly at small p ; the necessity of $d(r_\nu) < p$ and of $p \geq 7$ was confirmed by counterexample.
3. $b_a \leq 6\alpha x + 3$ (**p. 10**). True and sharp, but unproved in the paper and load-bearing for $L_a \geq 0$ at $x = 3$; see Finding 1. Two lines would fix it.
4. **The $m_N \geq 1$ part of the inner range.** For $M = 200$ this is most of the inner range and it has no determinant test at any K . The bookkeeping is uniform in m_N and was verified symbolically and over sweeps reaching $m_N = 14$.
5. $A_{200} + \bar{U} < 0$. Verified exactly, *given* γ_p^{in} and γ_p^{out} . The margin is 1.29%; an error of that size in either constant kills Theorem 1.1.

A structural remark that should reassure the author and the reader alike. By the paper’s own C.3, $Q_{K,M} = a_{K,M}P_K$ with $a_{K,M}$ a positive integer and $P_K = \Delta_K/\text{cont}(\Delta_K)$ primitive. Hence Theorem 1.1 follows from

$$\limsup_{K \rightarrow \infty, 40|K} K^{-2} \log P_K(\zeta(5)) < 0$$

alone, with no reference to $m_{K,M}$, to γ_p^{in} , or to Sections 3–5 at all: $P_K \in \mathbb{Z}[X]$ has degree h by (2.9), is positive at $\zeta(5)$ by Proposition 2.2, and the same final step applies. The entire arithmetic apparatus exists only to *prove* such a bound, by exhibiting an explicit divisor $m_{K,M}$ of the content. The measured values in Table 3 say that the quantity above is about -0.12 at computable K and extrapolates to -0.072 . So an error in Sections 3–5 would be an error in the

proof and quite probably not in the theorem. We say this not to excuse a gap but to locate it: a referee’s report on this paper should be about whether the denominators are proved, not about whether the construction works.

6 Minor and presentational remarks

These are offered to the author as corrections; none affects the mathematics.

1. (§2.) “Polynomial division and simple partial fractions extend μ_X ” (p. 3) is a definition on a basis, not an extension; the exclusion of $t = 0$ from the pole set is used — it is what makes Proposition 2.2’s integral converge — but never stated. In Proposition 2.2 (p. 5) the interchange of $\sum_{\ell \geq 1}$ and $\int_0^\infty$ is not justified (Tonelli, one line). In §2.3 (p. 4) the hedge “the *possible* poles $-j^2$ ” is confusing, since all h residues are nonzero, which is what makes V square and (2.9) nonvanishing; and $W = D_N^5$ is defined there but first used in §4.
2. (§3.) On p. 6 the citation “[9, (24.4.1), (24.4.3)]” appears to list the two DLMF displays in the opposite order to the two identities named in the sentence. We were not able to confirm the DLMF numbering (no offline copy; see Section 7), and in any case a bracketed list of two references carries no ordering, so this is at most a readability point. Lemma 3.3’s residue sign is left as “ $\pm$ ”; it is $(-1)^{K-r}$ uniformly. In Lemma 3.1 the hypothesis $\deg U_0 \leq p + 1$ may be relaxed to $4p - 2$ and $r_\nu - r_\eta \in \mathbb{Z}_p^\times$ is redundant — unused slack, worth knowing if §§4–5 ever needs it.
3. (pp. 10, (4.2)–(4.3).) The “ -4 ” and “ $+1$ ” silently absorb the p^{-4} prefactor of (3.7); the text does not say so, and settling it costs the reader a full re-derivation.
4. (p. 12.) The divided-difference integrality needs the $-\frac{1}{4} + \frac{1}{2j}$ tail of (2.3), via $a^4(H_{a-1}^{(5)} - H_a^{(5)}) + 1/a = 0$. The paper points at (2.3), so the reader has what is needed, but the one-line computation is missing; over 2087 node pairs the congruence holds 2087/2087 with the tail and 0/2087 without.
5. (p. 14, (5.7).) No explicit constant is given for the $O_M(1)$; the correct one is $3.8M$. “There are $O(M)$ such intervals” over-counts: in the z variable there are at most three pieces. The cancellation between the zero block and the ordinary block should be described as a cancellation rather than as an omission.
6. (p. 23, (A.6).) “The parenthesis is negative” follows in one line from the preceding sentence $\Phi'(\sqrt{q-}) < 0$ and the monotonicity of arctan; as printed it looks like an unsupported assertion, and it holds by only 2.5×10^{-8} .
7. (pp. 24–25, Table 2.) The caption explains only the single-run format. Rows with two runs are momentarily unreadable; one clause — that runs are separated by a space — would fix it.
8. (p. 26, (B.2).) Redundant: $2(1 - \alpha) = 2\lambda$ appears twice and the 2α -points are a subset of the 4α -points. Also, $H = \lambda + 3\alpha$ should be stated, since it is what makes the quadratic coefficient of R vanish identically.

9. (*Appendix C.*) The displayed $h^2\mathcal{R}^{2h}$ in (C.5) is true and reproducible but weaker by the $O(1)$ factor $\mathcal{R}^2 = 293.84$ than the same step gives. “Its positive margin absorbs the polynomial factor in h ” (p. 30) needs $K \gtrsim 2.81 \times 10^5$; neither the margin 1.433×10^{-4} nor the threshold is stated. And “the threshold b_0 is not specified” understates: b_0 is not *effective*, and is at least $\exp(1.07 \times 10^{11})$.
10. *Notation.* s is the number of subintervals in B.1, the quantity $Hx - T/2$ in (5.4), and appears in $O(K^{3/2})$; d is $d(r)$ in §3, $d(y)$ in (5.9) and d_0 in B.1; M is the cutoff while M_0 is a potential constant; N is $3n$ while $\mathcal{N}(x)$ is (5.5).
11. *Robustness (the most valuable single change).* Quoting (6.2) with $M_0 = -6.6498$ instead of -6.645 — which the author’s own data support — would multiply the final margin in (7.2) by 139 (from 0.0525 to 7.296) and remove the knife-edge from (6.4), (A.9) and the printed exponent at a stroke. Similarly, the sanity check that the machinery gives $+0.93$ for $\zeta(7)$ and -0.86 for $\zeta(3)$ is half a page and answers the first question every referee will ask.
12. “*Author’s note on further work*” (p. 31). A page of unsubstantiated claims of “several advances” and “substantial progress” with no mathematical content, which the section itself then disclaims. It raises a referee’s prior against the paper for no gain and should be deleted.
13. *Reproducibility.* The paper should ship an ancillary verification script. Everything in Appendices A and B is reproducible at ordinary precision in minutes — we did it — but a referee should not have to reimplement the rational-enclosure apparatus to check a lemma.

7 What we did not check, and the limits of this audit

7.1 Not checked

1. $v_p^G(\Delta_K) \geq \gamma_p^{\text{in}}$ **at admissible parameters.** The central omission. Hypothesis (4.1) forces $h \geq 296000$; the smallest admissible matrix is 296000×296000 . What would settle it: a machine-checked formalisation of §3 plus §4.1 (the argument is finite and elementary), or a targeted determinant over $\mathbb{Z}_p$ in the unimodular basis (4.5) at a single admissible prime, or — much the cheapest — a rewrite of §4 admitting small M (e.g. $M = 3$, $K \approx 1800$) so that the prime-by-prime comparison becomes computable.
2. **The $m_N \geq 1$ portion of the inner range**, verified only by symbolic bookkeeping.
3. **The author’s own exact-rational enclosure pipeline** over the 684 intervals of Table 2. We verified the same inequalities in 25–60-digit interval-free arithmetic against a smallest margin of 6.9×10^{-7} , and verified analytically that the prescribed truncation orders leave remainders 10^{37} times smaller than needed. The residual risk is negligible but not zero; an interval-arithmetic or formally verified rerun would close it.
4. **An independent transcription of Tables 1 and 2.** Ours checksums (exact mass $37/40$; exact tiling of $[0, 2]$), which is strong evidence, but a second pair of eyes on Table 1 is worth one referee-hour given the 6.45×10^{-8} margin in (6.4).

5. **Every proof step that is not a computation.** The audit is a machine audit with a line-by-line reading on top of it. The proofs of **Lemmas 3.1, 3.2, 3.3**, of **Lemma 4.2**, of **Lemma 6.2**, and the partial-summation step inside **Proposition 5.2** were re-derived by hand and then tested only through their *consequences* (exactly, at small p and small K , and by randomised stress tests). None of them has been verified as a theorem at the parameters §§4–5 use them. We found no defect in any of them, and Lemma 4.2’s statement is a finite claim about $h \times h$ matrices that 6000 random instances did not break; but a machine-checked proof is the thing that would settle them.
6. **The degree bound on p. 9** (“the pullback numerator has degree at most $12N + 4h + 1$ ”), an input to Lemma 3.3 and hence to (3.12). We confirmed the *use* made of it — $\max(2K, d + 1) = 184n + 2 \leq 200n = 5K$, and replacing $\max(2K, d + 1)$ by $5K$ weakens (3.10), which is the safe direction — but not the bound itself.
7. **The DLMF equation numbers** cited on pp. 5, 6 and 29 ((25.6.2), (25.11.29), (24.4.1), (24.4.3), §24.10(i), (18.10.5)). We verified the *identities* numerically (Euler’s $\zeta(2m)$ formula symbolically; Hermite’s formula to ≥ 24 digits; von Staudt–Clausen through $v_p(\kappa_d) \geq -1$; the Legendre bound $|P_j(s)| \leq R^j$ for $j \leq 24$); we did not open DLMF to confirm that those are their numbers.
8. **The (α, c) optimisation behind the $\zeta(s)$ sanity test.** The table of balances at the paper’s own $(\alpha, \lambda) = (3/40, 37/40)$ was reproduced exactly by two tracks; the further claim that re-optimising (α, c) still leaves $\zeta(7)$ failing (+0.84) rests on one track’s computation and was not independently re-run.
9. **Whether a known no-go result applies.** Zudilin’s $\mathbb{Z}$ -transfinite-diameter obstruction, F. Brown’s recent bound and the Chudnovsky–Bombieri G -function bounds all constrain linear-form or Mellin constructions; none that we found applies verbatim to a determinant producing integer *polynomials* Q with $Q(\zeta(5))$ small. A short answer from a specialist (Zudilin, Brown, Fischler) is the one external input that could settle the paper quickly in either direction. Separately, the author and this construction appear nowhere we could find. The bibliography [1]–[13] is a list of real papers; we checked four of them at source ([4] Brown–Zudilin, [5] Fischler, [6] Fischler–Sprang–Zudilin, [8] Lai–Lupu–Sprang) and the paper characterises each accurately, including the two quotations on p. 1.

7.2 Where the audit corrected itself

Discipline requires recording where our own first pass was wrong, since several of these would have become false accusations against the paper.

- **Table 3, row $j = 14$ (p. 27).** One verification pass reported that the paper prints 167757646109/955086612480 where the truth is 16775764609/955086612480, and called it “the only outright false printed statement in the paper”. **This is wrong, and we settled it ourselves.** The PDF text layer reads 16775764609, and so does a 600 dpi render of p. 27, where the numerator is unambiguously eleven digits long; it is only at screen resolution that the glyphs invited a misread, which two separate passes then made in the same way. The decisive test is arithmetic rather than typographic: only 16775764609/955086612480 makes the 17 printed rows sum to (5.18), and with it the sum is *exactly* 322437603634266857629/7535670527041937280000 (recomputed here in **Fraction**

arithmetic). **The paper is correct here, and its sentence “Grouping its terms into unit intervals gives Table 3. Their sum is (5.18)” is true as printed.** General rule for anyone re-doing this work: take digits from the text layer or from a high-resolution render, never from a screen-resolution page image.

- **The (A.9) margin.** One track reported 6.892×10^{-10} ; its verifier reported 6.892×10^{-7} . We recomputed it from our own parse of Table 2: the margin is 6.8921559×10^{-7} . The verifier is right.
- **“Table 2 is unparseable, so (A.9) cannot be checked”** (recorded as a major finding on the first pass). Refuted: the k -column lists whitespace-separated runs, the 684 intervals tile $[0, 2]$ exactly, and (A.9) holds on all of them. Lemma 6.1 moves from “could not check” to *verified*. Only the caption needs a clause.
- **“The Bernoulli convention $B_1 = -1/2$ is never stated.”** False: it is stated on p. 3, one line before (2.2). We confirmed this in the text layer.
- **“ $p^2 > 5K$ is asserted but never used.”** False: it is used on p. 14 in §5.1, to truncate Legendre’s formula (5.3) to its $a = 1$ terms.
- **“Proposition 4.1 relies on three side conditions never displayed.”** False: all three are displayed in the proof on pp. 10–11. They are consequences of (4.1) proved inside the proof, not hidden assumptions. (The genuine gap in that stretch is the separate $b_a \leq 6\alpha x + 3$, Finding 1.)
- **“pdftotext garbles (4.9) as $5K \leq 2p - 2$.”** False for our text extraction: line 613 of it reads $5N \leq 2p - 2$, matching the PDF. Publishing this caution would have sent readers chasing a non-existent problem.
- **“(3.8)’s proof suppresses a case split.”** A misreading: the paper proves (3.8) by Vandermonde together with $\binom{z}{j} = \frac{z}{j} \binom{z-1}{j-1}$, which needs no case split.
- **The uniformity constant in (5.7).** One track reported $0 < \gamma_p^{\text{in}} - p\Gamma < 7.41M^2$, always positive; its verifier traced this to a factor-2 bug in that track’s code (a cap on the zero-block weights applied to an already-doubled quantity) and obtained $|\gamma_p^{\text{in}} - p\Gamma| \leq 3.8M$ with both signs. The verifier reproduced the erroneous numbers with the buggy line and the correct ones without it. The paper is unaffected, and is in fact better served by the corrected statement.
- **Smaller corrections** adopted from verification passes: the sharp constant in (6.11) is $(2\pi)^4/12 = 129.88$, not 123.575 (a grid artefact); $C_p \in p^5\mathbb{Z}_p^\times$ is false at $p = 37$, the correct statement being $v_p(C_p) \geq 5$ with equality unless $p \mid B_{p-5}$; Proposition 4.3’s bound is attained at 27 of 37 primes, not 26; $u > 0$ occurs already at $K = 80$; and (C.5)’s finding was downgraded to cosmetic once it was seen that the paper’s displayed form *is* reproducible from its own argument.

7.3 Limits

This audit is numerical and structural, not a formal verification. Its strongest claims are of the form “this identity is exact” and “this formula is exactly attained at 27 independent primes”; its

weakest concern the asymptotic regime the paper actually inhabits ($K \geq 8 \times 10^6$ for $M = 200$, $K \geq 2 \times 10^{12}$ for Corollary 1.2). Every extrapolation we report — $\overline{U}_{\text{true}}$, the decay of $P_K(\zeta(5))$, the constant A_∞ — is a fit over $K \leq 240$, validated out of sample where possible but not proved; and where the paper’s statements are limsups with no effective threshold, no finite computation can confirm them, so we checked the constants they are made of instead. What the audit did *not* consist of is reading the paper for plausibility: every track reproduced a known answer — Euler’s $\zeta(2m)$, Hilbert determinants, Marchenko–Pastur, von Staudt–Clausen, Raabe, DLMF 25.11.29 — before it was allowed a verdict, and was then attacked by a second independent implementation. In every case, the paper survived.

Provenance and authorship

This report was written by **Claude Opus 5 (1M context)** (model identifier `claude-opus-5[1m]`), an AI system made by Anthropic, running in Claude Code on Dan Romik’s machine on 22 September 2026. No part of it has been refereed by a human mathematician. It is signed so that a reader can weigh it accordingly, and so that any error in it is attributable.

How it was produced. The audit ran as an orchestrated workflow of twenty agent instances of the same model: nine audit tracks — the construction and Appendix C; the local functional (§3); the local determinant estimates (§4); normalization and the prime sum (§5, Appendix B); the real bound (§6, Appendix A); exact computation of $\Delta_K(X)$; independent recomputation of the two constants; a first-principles equilibrium analysis; and the strategy, logic and literature — each followed by a *separate* adversarial verification pass instructed to attack the track’s findings from both sides, re-implementing the decisive computations by a different method; then a writer and a fact-checker. Each track was required to reproduce a known answer before it was permitted a verdict. The verification passes materially changed the report: they withdrew two of the §4 findings as mistaken, moved Lemma 6.1 from “could not check” to verified, and refuted a claimed transcription error in Table 3 by rendering the page at 600 dpi. The fact-checking pass found twenty further defects in the draft, five of them substantive, including two wrong final digits in A_{200} and A_{100000} and a confidence figure higher than the weakest component it aggregated.